

# **Spatial Machine Learning in R**

Degree of Urbanisation (DEGURBA)  
urban structure as a spatial feature

Session 4

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# Why urban structure matters in spatial models

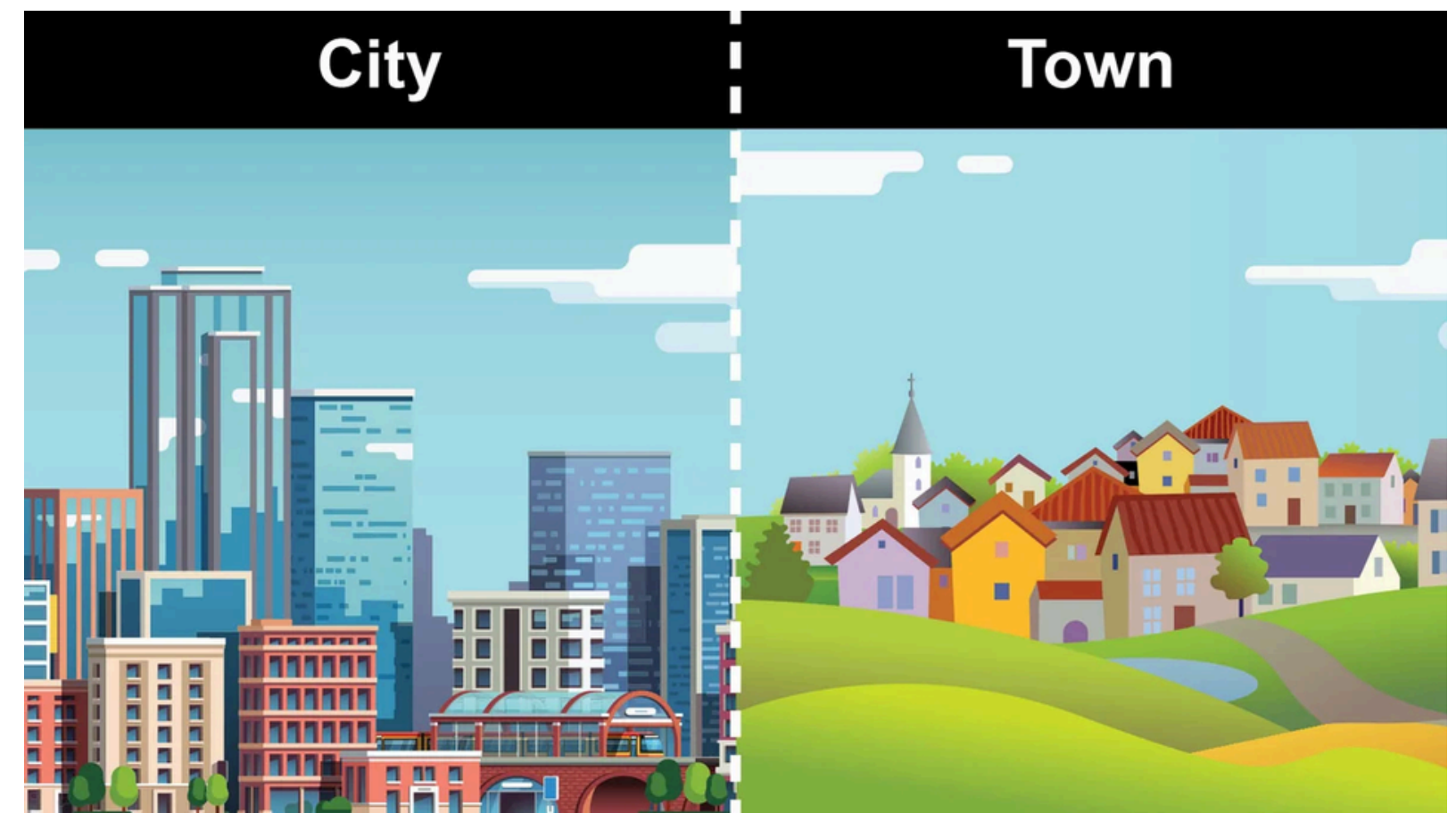
Many spatial processes depend strongly on the structure of **human settlements**.

Cities, suburbs, and rural areas differ in population density, infrastructure availability, and economic activity. Because of these differences, spatial outcomes often follow a clear **urban–rural gradient**.

## Examples:

- energy consumption
- mobility patterns
- housing prices
- pollution exposure
- healthcare accessibility

**In spatial modelling, capturing these structural differences is important because they influence how spatial relationships behave across space!**



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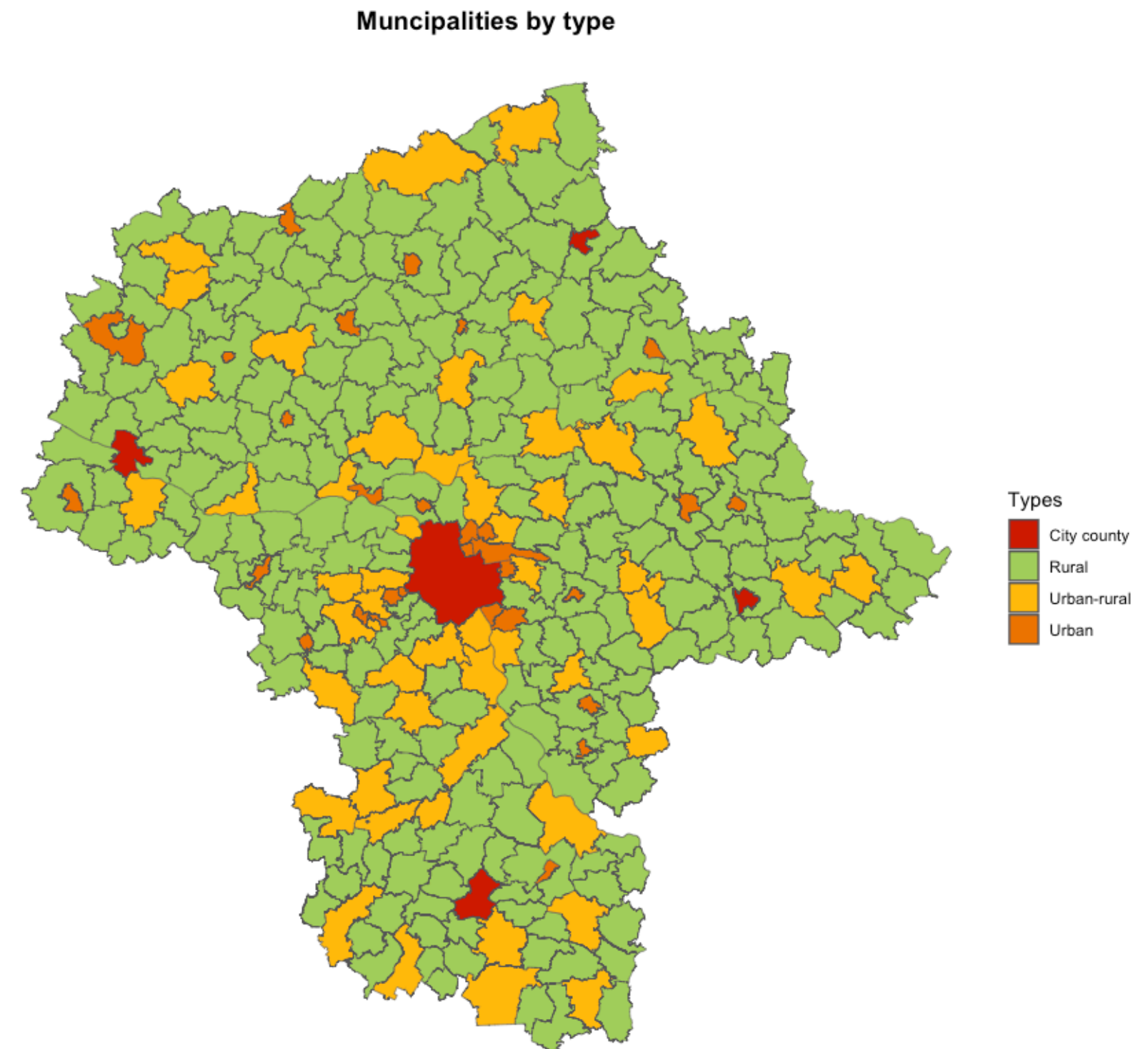
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# The problem with administrative definitions of cities

## Administrative boundaries $\neq$ urban structure

A single administrative unit may contain both dense urban areas and large rural territories. At the same time, suburban areas that function as part of a city may lie outside the official city boundary.

Because **administrative systems differ between countries**, using these units to represent urbanisation creates problems for **comparability and spatial analysis**.



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# Population grids as a spatial representation

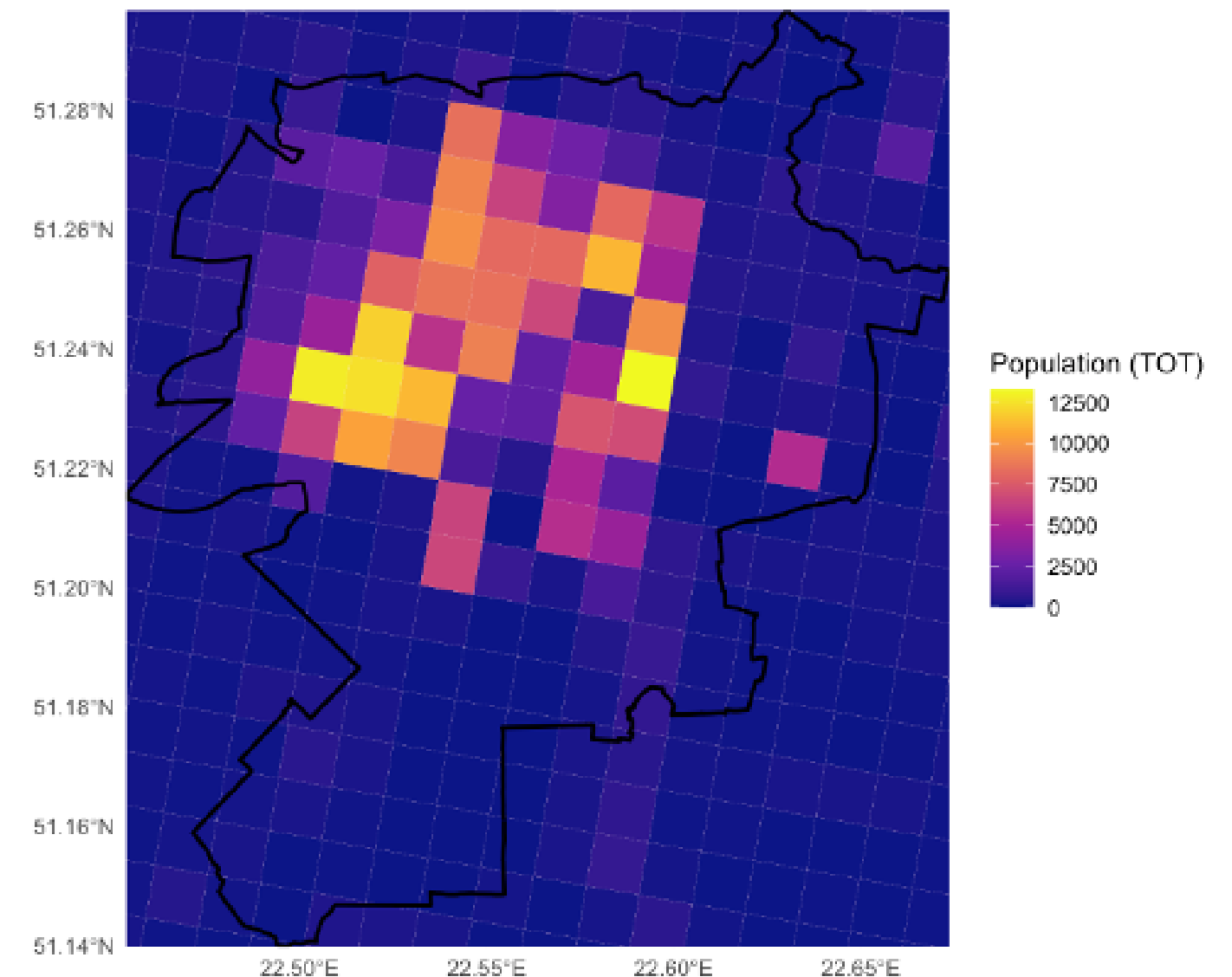
**Solution: represent population continuously in space**

To better capture settlement patterns, the degree of urbanisation methodology uses population grids instead of administrative units.

## Population grid:

- 1 km × 1 km cells
- each cell contains population count

Using a grid-based representation allows researchers to detect **spatial clusters** of population density and identify urban centres independently of administrative boundaries.





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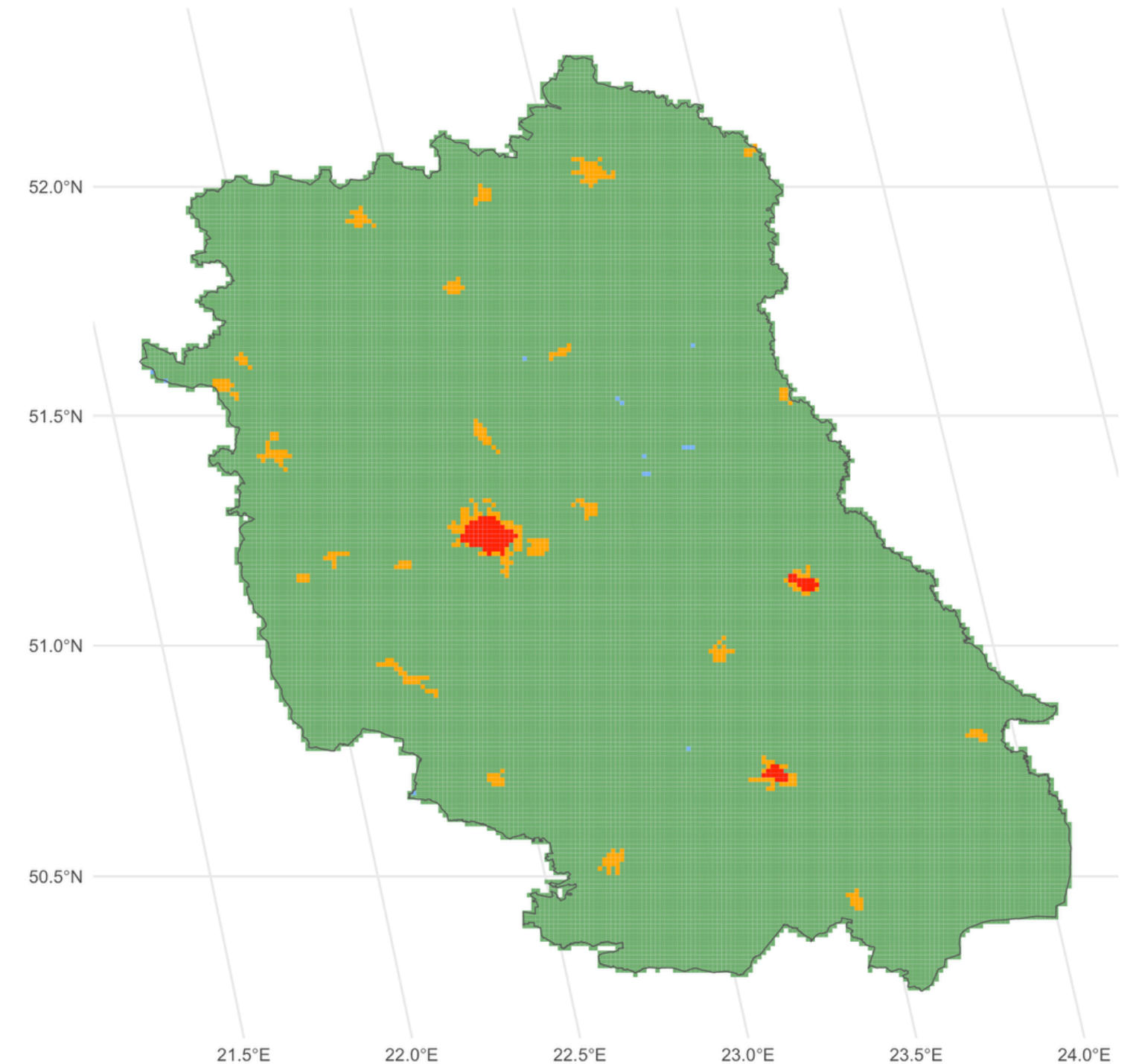
# DEGURBA as a spatial classification

The **Degree of Urbanisation (DEGURBA)** classification identifies different types of settlements based on population density and spatial clustering of the population grids.

The methodology distinguishes three main categories at Level 1:

- **Cities** (densely populated areas)
- **Towns and semi-dense areas**
- **Rural areas** (thinly populated areas)

These categories capture the continuum between urban and rural environments, providing a consistent way to analyse settlement structure across countries.



# Detecting population clusters

The crucial step is to classify the grid cells according to the number of people and group neighbouring grid cells into clusters.

## Urban centre (high-density clusters)

- $\geq 1500$  people/km<sup>2</sup>
- $\geq 50,000$  population

## Urban cluster (mid-density clusters)

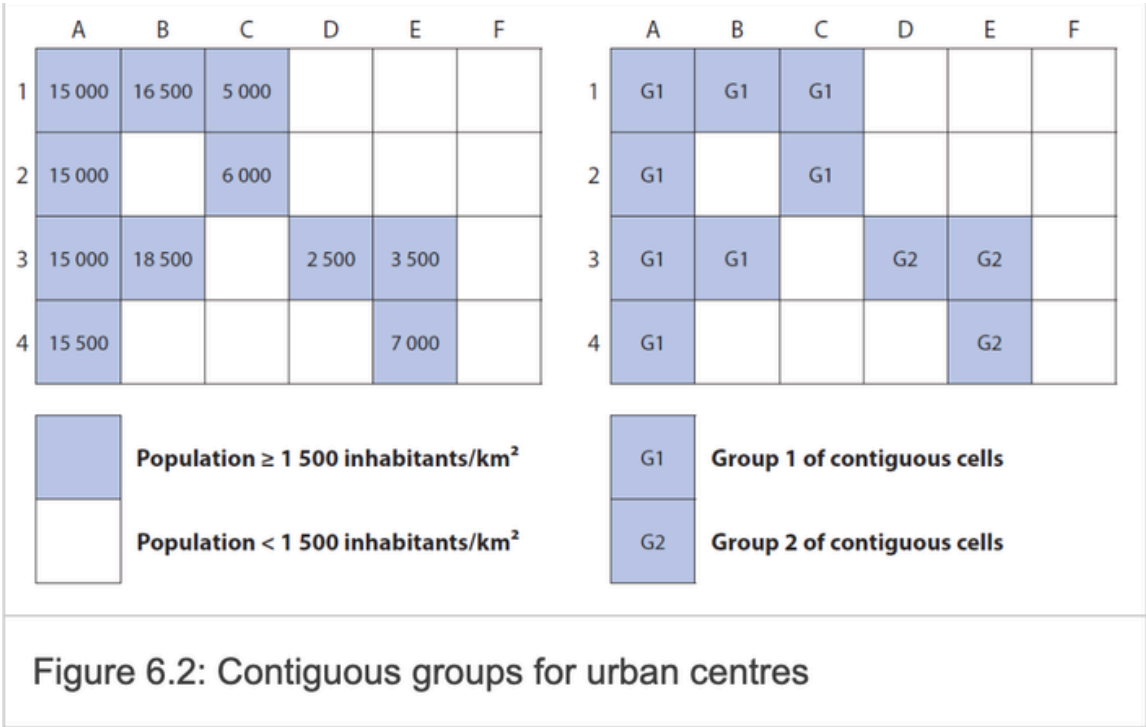
- $\geq 300$  people/km<sup>2</sup>
- $\geq 5,000$  population

## Rural cells (low-density cells)

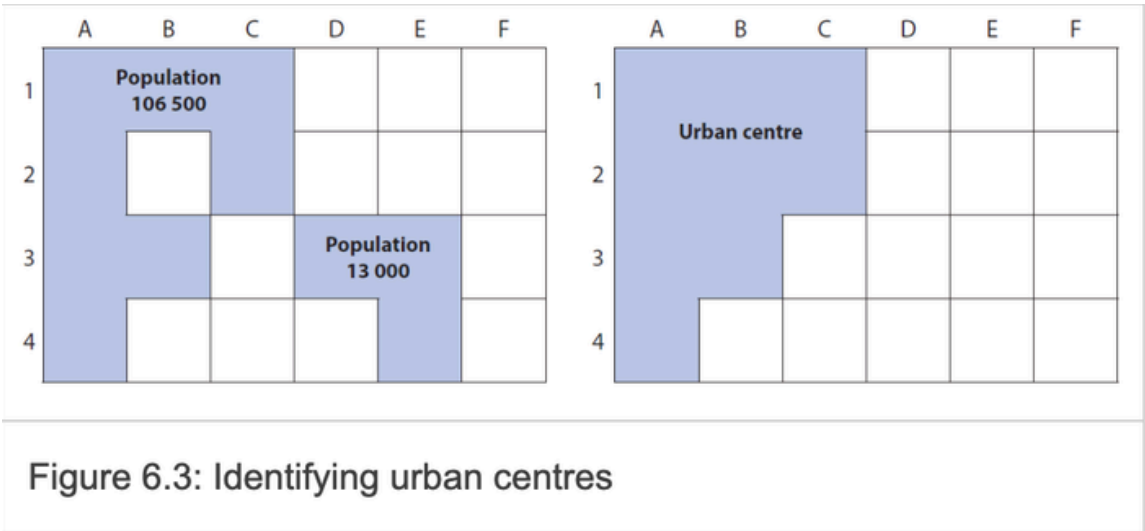
- all other cells

Table 5.3: Degree of urbanisation (DEGURBA): Level 1

| L1                                                             |               | Population size thresholds of the cluster of cells |               | No population size criterion |
|----------------------------------------------------------------|---------------|----------------------------------------------------|---------------|------------------------------|
|                                                                |               | $\geq 50\,000$                                     | $\geq 5\,000$ |                              |
| Population density of cells (inhabitants per km <sup>2</sup> ) | $\geq 1\,500$ | Urban centre                                       | Urban cluster | Rural grid cell              |
|                                                                | $\geq 900$    |                                                    |               |                              |
|                                                                | $\geq 300$    |                                                    |               |                              |
|                                                                | $\geq 50$     |                                                    |               |                              |
|                                                                | $> 0$         |                                                    |               |                              |



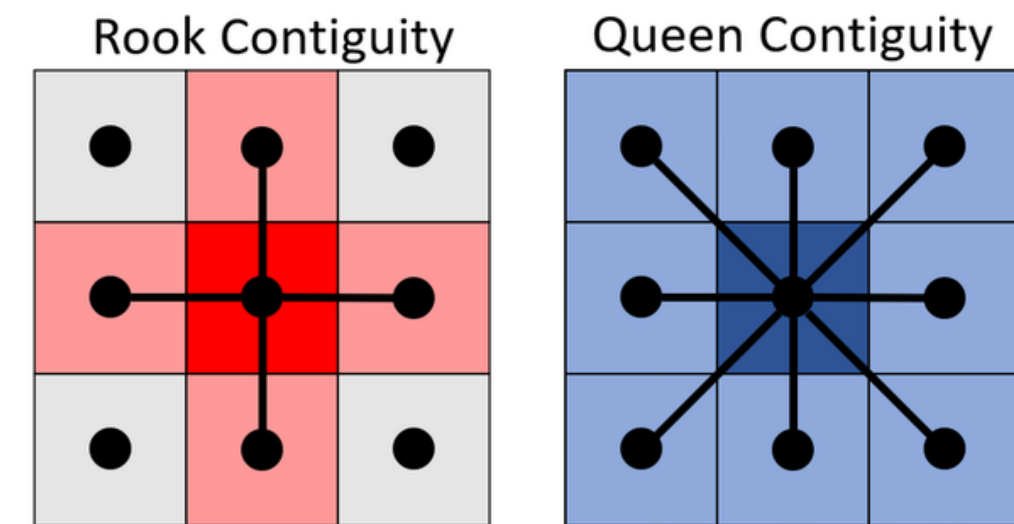
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# Neighbour definition

- A key step is defining **spatial neighbourhoods between grid cells**
- **Urban centres:** rook contiguity (cells sharing an edge)
- **Urban clusters:** queen contiguity (cells sharing an edge or corner)
- This choice influences **cluster shape and connectivity**



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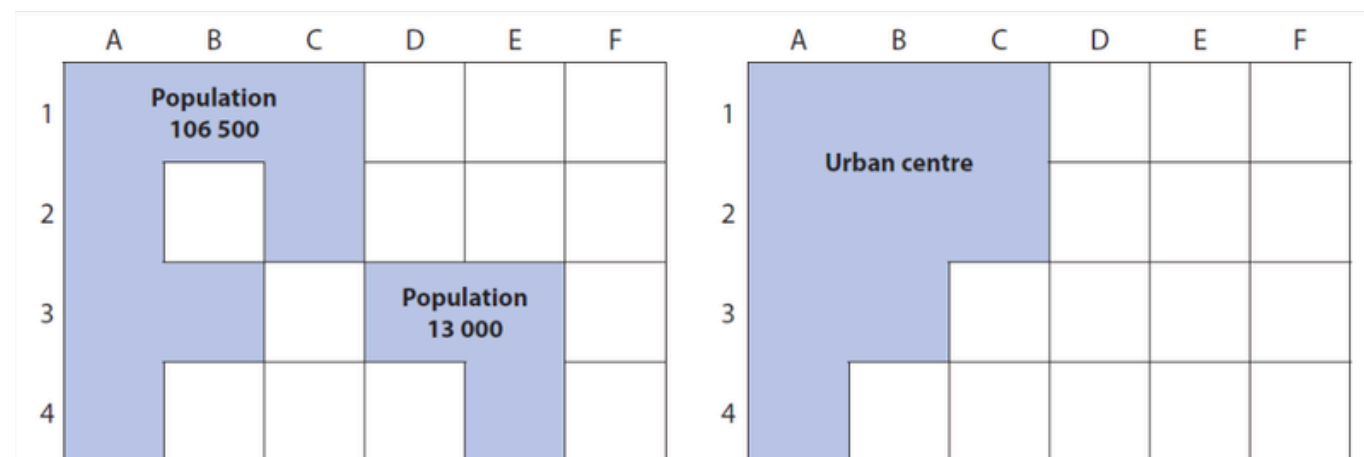


Figure 6.3: Identifying urban centres

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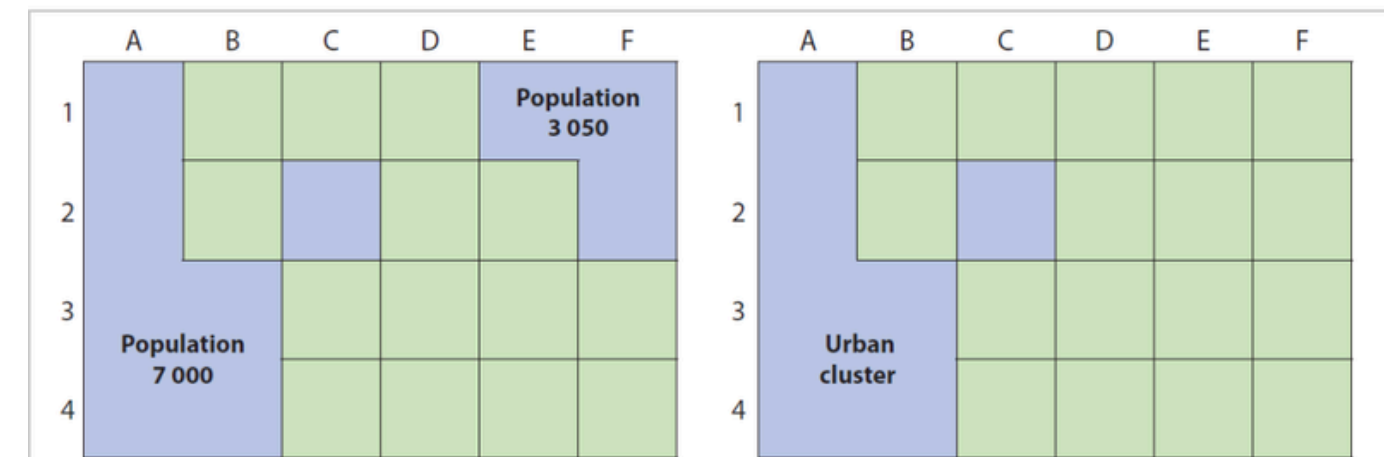


Figure 6.5: Identifying urban clusters

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# Level 2 DEGURBA: a more detailed settlement hierarchy

## Level 1 classification:

- Cities
- Towns and semi-dense areas
- Rural areas

## Level 2 provides more detail by splitting these categories into:

### Towns and semi-dense areas

- Dense towns
- Semi-dense towns
- Suburban / peri-urban areas

### Rural areas

- Villages
- Dispersed rural areas
- Mostly uninhabited areas

| L1     | Minimum population size of the cluster of cells<br>(settlement size) |               |                 |
|--------|----------------------------------------------------------------------|---------------|-----------------|
|        | ≥50 000                                                              | ≥5 000        | none            |
| ≥1 500 | Urban centre                                                         | Urban cluster | Rural grid cell |
| ≥900   |                                                                      |               |                 |
| ≥300   |                                                                      |               |                 |
| ≥50    |                                                                      |               |                 |
| >0     |                                                                      |               |                 |

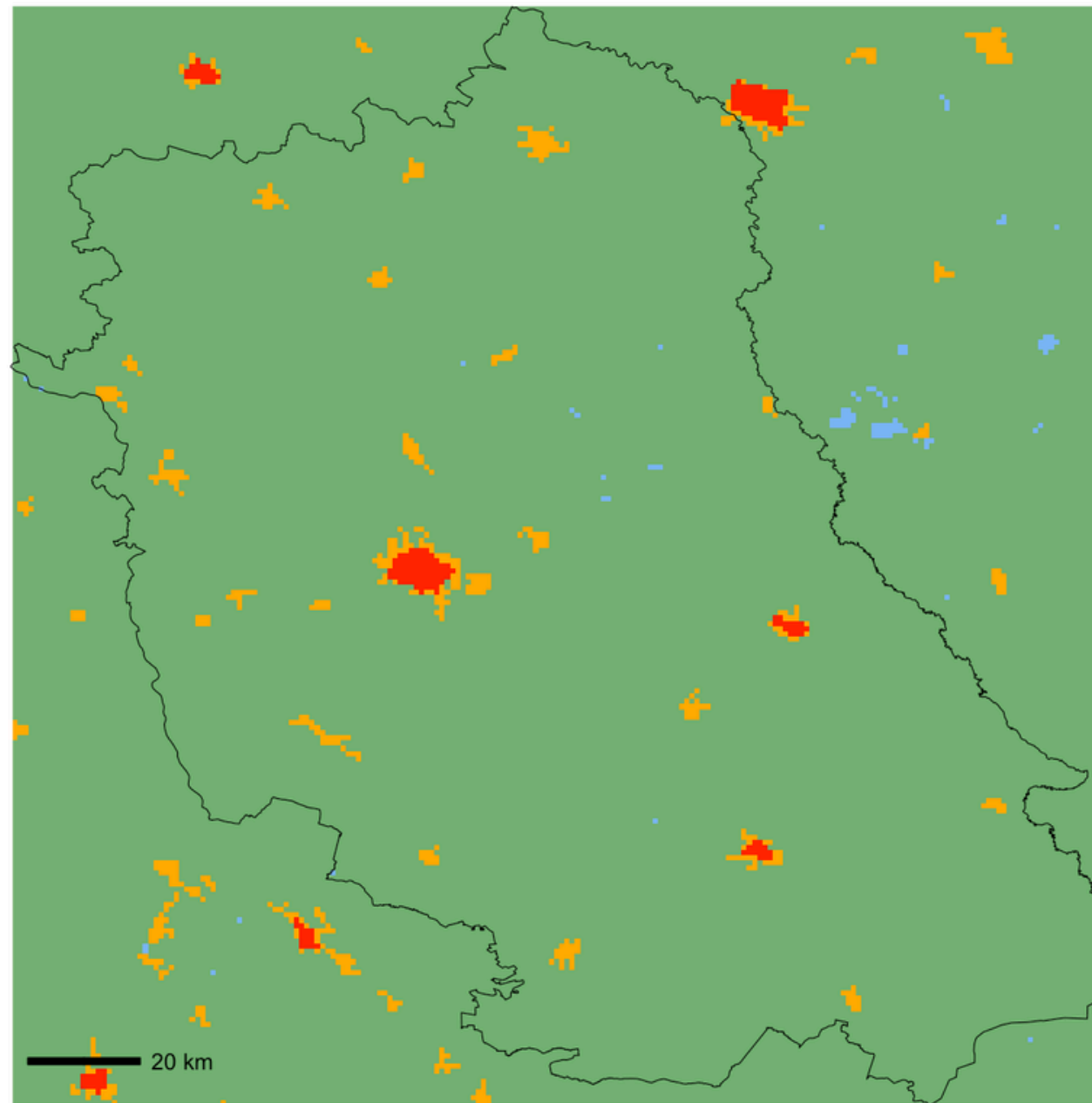
| L2     | Minimum population size of the cluster of cells<br>(settlement size) |                                  |                          |                                  |                            |                            |
|--------|----------------------------------------------------------------------|----------------------------------|--------------------------|----------------------------------|----------------------------|----------------------------|
|        | ≥50 000                                                              | ≥5 000                           | ≥2 500                   | none                             | ≥500                       | none                       |
| ≥1 500 | Urban centre                                                         | Dense urban cluster <sup>1</sup> | Semi-dense urban cluster | Suburban or peri-urban grid cell | Rural cluster <sup>1</sup> | Low density grid cell      |
| ≥900   |                                                                      |                                  |                          |                                  |                            |                            |
| ≥300   |                                                                      |                                  |                          |                                  |                            |                            |
| ≥50    |                                                                      |                                  |                          |                                  |                            | Very low density grid cell |
| >0     |                                                                      |                                  |                          |                                  |                            |                            |

<sup>1</sup>Dense urban clusters and Rural clusters have an upper population size bound at 50 000 and 5 000 respectively



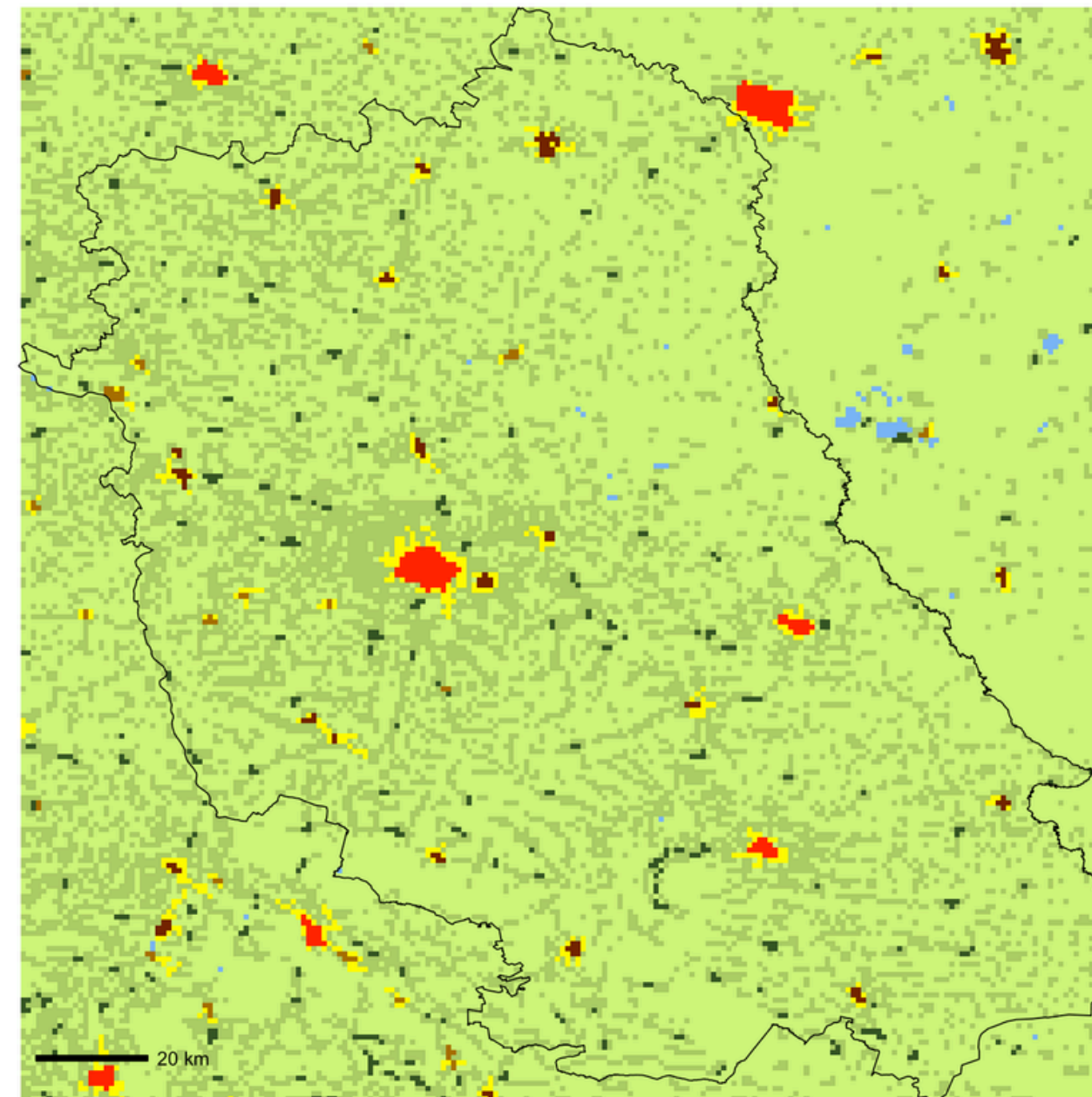
# Example of Level 1 vs. Level 2 classification

DEGURBA L1 for Lubelskie



urban centre urban cluster rural cell water cell

DEGURBA L2 for Lubelskie



urban centre dense urban cluster semi-dense urban cluster suburban or peri-urban cell rural cluster low density rural cell very low density rural cell water cell

# Additional conditions checked in the algorithm

Beyond population density, several geographic and methodological checks are applied

## Built-up areas

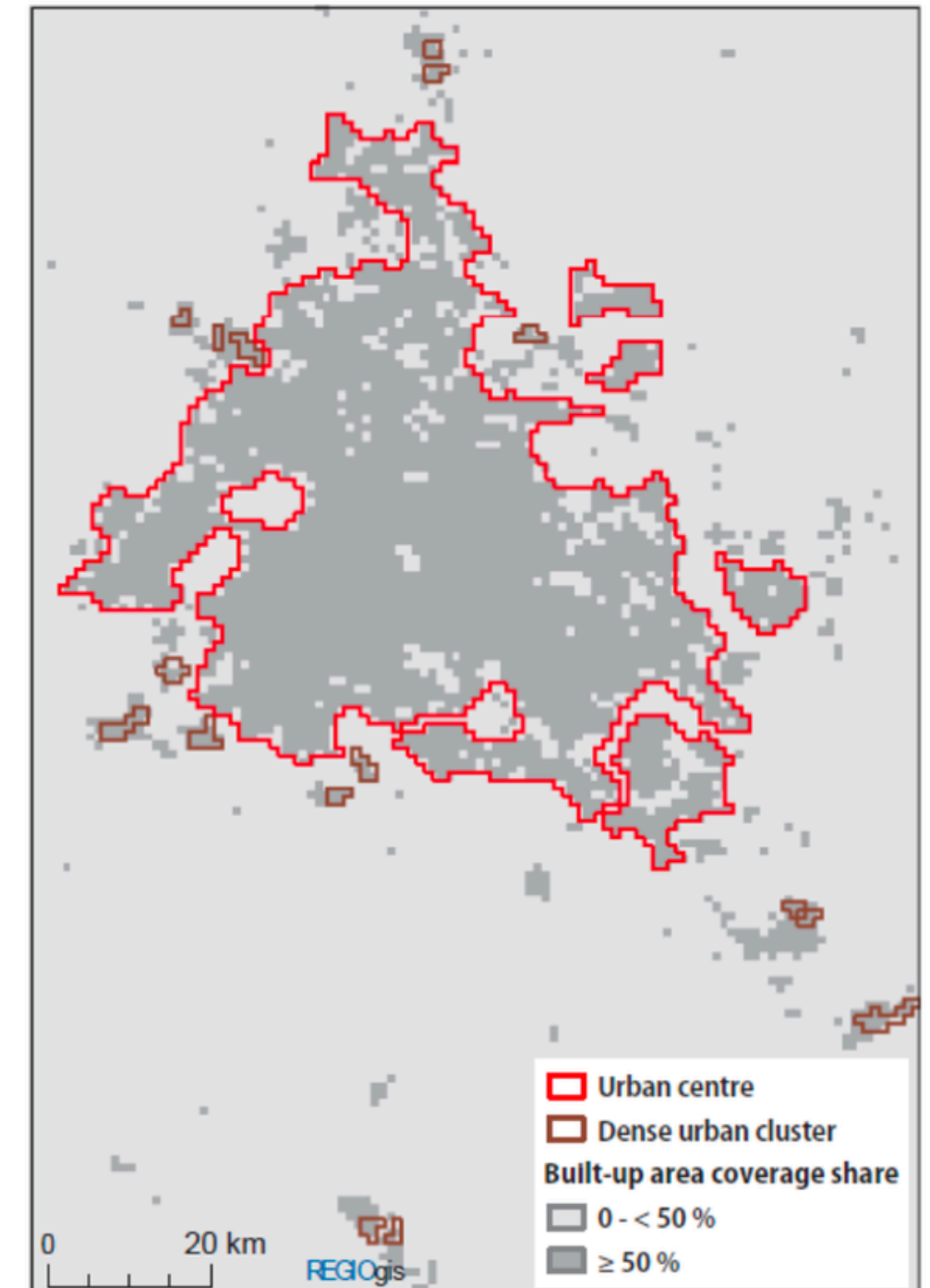
- Areas with buildings, roads, or infrastructure may have low residential population
- Built-up data can be used to connect fragmented urban cells
- Prevents transport corridors, industrial zones, or commercial areas from splitting cities

## Water bodies

- Rivers, lakes, and coastlines may create artificial gaps in population grids
- Adjustments allow clusters separated by water to be evaluated properly

## Gap filling and smoothing

- Small holes inside dense clusters are filled
- Borders of clusters are smoothed to create realistic urban shapes



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# Why this classification is useful in spatial modelling

DEGURBA can be a spatial feature in models!

The classification can be used to:

- describe the **urban–rural context** of observations
- include settlement structure as an **explanatory variable**
- capture **spatial heterogeneity** in models

Examples in spatial modelling:

- energy demand models
- accessibility modelling
- housing price prediction

